

Distributed Normalized Maximum Correntropy Algorithm with Step-Size Control

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Abstract—In this work, we investigate robust distributed adaptive algorithms based on the maximum correntropy criterion (MCC) for distributed estimation problems. To address the performance degradation of the diffusion MCC (DMCC) algorithm under varying input signal powers, we propose a diffusion normalized MCC (DNMCC) algorithm. Furthermore, to resolve the trade-off between convergence speed and steady-state accuracy, a step-size control (SSC) mechanism is developed for DNMCC, leading to the SSC-DNMCC algorithm. The SSC strategy selects a suitable step-size from a predefined decreasing set based on a dynamic MSD-trend prediction model. Simulation results for a sensor network verify the effectiveness of the proposed algorithms compared with existing methods.

Index Terms—Diffusion, Impulsive noises, Maximum correntropy criterion, Mean square deviation, Step-size control.

I. INTRODUCTION

In distributed parameter estimation, network nodes cooperatively estimate unknown parameters through local processing and neighborhood information exchange, thereby avoiding the single-point failure inherent in centralized schemes and offering strong robustness against node or link disruptions [1]. Owing to these advantages, distributed algorithms have been widely applied in power spectrum estimation, acoustic localization, and biomedical signal analysis [2]–[4]. Among them, diffusion-type strategies [5] have attracted particular attention due to their adaptability to topology variations and their performance comparable to centralized approaches [6]. Based on the mean square error (MSE) criterion, classical diffusion algorithms, such as the diffusion least mean square (DLMS) algorithm [7] and the diffusion normalized least mean square (DNLMS) algorithm [8], have been developed; however, they remain highly sensitive to impulsive disturbances commonly encountered in practical environments.

To enhance robustness under impulsive noises, various different from the MSE cost functions have been proposed. The p -norm error criterion gives rise to methods such as the diffusion least mean p -power [9] and diffusion continuous mixed p -norm [10] algorithms, but their performances depend critically on the choice of p . The information theoretic learning (ITL) framework [11] offers another pathway for handling impulsive disturbances. To name a few, the diffusion minimum error entropy (MEE) algorithm [12] minimizes the entropy of recent error samples but incurs high computational com-

plexity because of its double-summation form. In contrast, the maximum correntropy criterion (MCC) provides strong outlier suppression with significantly lower complexity. The diffusion MCC (DMCC) algorithm [13] leverages this property, yet its performance remains sensitive to input signal power variations.

Despite ITL-based algorithms having robustness to impulsive noises, they still rely on fixed step-sizes, leading to a fundamental trade-off between convergence speed and steady-state accuracy. To alleviate this limitation, numerous variable step-size (VSS) strategies have been proposed in Gaussian environments [14], [15], and later extended to non-Gaussian scenarios. For example, the generalized VSS diffusion continuous mixed p -norm (GVSS-DCMPN) algorithm improves robustness but it requires careful parameter tuning [10]. Other works employ M-estimation principles to design the diffusion robust VSS LMS [16] and VSS affine projection algorithms [17].

To resolve the step-size constraint inherent in the DMCC algorithm, this work presents a step-size control (SSC) concept to improve convergence and steady-state performances. We first introduce a diffusion normalized MCC (DNMCC) algorithm to reduce the sensitivity of DMCC to input power variations. Moreover, a practical mean-square-deviation (MSD) prediction formula is established to characterize the descending trend of the MSD under different step-sizes. This facilitates the design of an SSC mechanism that selects the step-size yielding the most favorable predicted MSD decrease, leading to the proposed SSC-DNMCC algorithm.

The remainder of this paper is organized as follows. Section II provides a brief review of the DMCC algorithm. Section III develops the SSC-DNMCC algorithms. The effectiveness of the proposed SSC-DNMCC algorithm is demonstrated through simulations in Section IV. Finally, Section V concludes the paper.

II. SIGNAL MODEL AND PROBLEM STATEMENT

A distributed network consisting of N sensor nodes in different spatial locations is considered, where at each node k and at each time i , an M -dimensional input signal vector $\mathbf{u}_k(i)$ and a desired signal $d_k(i)$ satisfy the following linear model:

$$d_k(i) = \mathbf{u}_k^T(i) \mathbf{w}_o + v_k(i), \quad (1)$$

where $\mathbf{w}_o$ represents an unknown vector of length M to be estimated, and $v_k(i)$ denotes the additive noise.

Considering the impulsiveness of $v_k(i)$, a widely-used choice for the kernel function in the correntropy estimate is the Gaussian kernel, which assigns larger weights to small errors while effectively suppressing the influence of large deviations caused by impulsive noises. It is defined as [18]

$$g(e_k(i)) = \exp\left(-\frac{e_k^2(i)}{2\sigma^2}\right), \quad (2)$$

where

$$e_k(i) = d_k(i) - \mathbf{u}_k^T(i)\mathbf{w}_k, \quad (3)$$

with $\mathbf{w}_k$ representing an estimate of $\mathbf{w}_o$ at node k and σ denoting the kernel width.

In diffusion-based distributed algorithms, each iteration at every node consists of an adaptation step followed by a combination step. For the DMCC algorithm [13], the adaptation step computes the intermediate estimate $\psi_k(i)$ according to

$$\psi_k(i) = \mathbf{w}_k(i-1) + \eta_k g(e_k(i)) e_k(i) \mathbf{u}_k(i), \quad (4)$$

where η_k is the step-size which controls the dynamic behavior of the algorithm. In the combination step, each node fuses intermediate estimates $\psi_l(i)$ from its neighbor nodes to obtain a more reliable estimate:

$$\mathbf{w}_k(i) = \sum_{l \in N_k} c_{l,k} \psi_l(i), \quad (5)$$

where $\{c_{l,k}\}$ are combining coefficients in the combination step, with the property that $\sum_{l \in N_k} c_{l,k} = 1$ and $c_{l,k} \geq 0$ [19].

III. PROPOSED SSC-DNMCC ALGORITHM

Although the DMCC algorithm exhibits robust performance in non-Gaussian impulsive noise environments, its weight update mechanism remains susceptible to variations in the input signal power. To alleviate this sensitivity, we normalize the adaptation step by the input vector energy and obtain equations of DNMCC as

$$\psi_k(i) = \mathbf{w}_k(i-1) + \eta_k \frac{g(e_k(i)) e_k(i) \mathbf{u}_k(i)}{\|\mathbf{u}_k(i)\|_2^2}, \quad (6a)$$

$$\mathbf{w}_k(i) = \sum_{l \in N_k} c_{l,k} \psi_l(i). \quad (6b)$$

Denoting the weight deviation vectors by $\tilde{\psi}_k(i) = \mathbf{w}_o - \psi_k(i)$ and $\tilde{\mathbf{w}}_k(i) = \mathbf{w}_o - \mathbf{w}_k(i)$, then subtracting (6a) from $\mathbf{w}_o$ yields

$$\tilde{\psi}_k(i) = \tilde{\mathbf{w}}_k(i-1) - \eta_k \frac{g(e_k(i)) e_k(i) \mathbf{u}_k(i)}{\|\mathbf{u}_k(i)\|_2^2}. \quad (7)$$

In accordance with (1) and (3), the error signal $e_k(i)$ can be rewritten as

$$e_k(i) = \mathbf{u}_k^T(i) \tilde{\mathbf{w}}_k(i-1) + v_k(i). \quad (8)$$

Substituting (8) into (7), it yields

$$\tilde{\psi}_k(i) = \mathbf{F}_k(i) \tilde{\mathbf{w}}_k(i-1) - \eta_k g(e_k(i)) \frac{v_k(i) \mathbf{u}_k(i)}{\|\mathbf{u}_k(i)\|_2^2}, \quad (9)$$

where

$$\mathbf{F}_k(i) = \mathbf{I} - \eta_k g(e_k(i)) \frac{\mathbf{u}_k(i) \mathbf{u}_k^T(i)}{\|\mathbf{u}_k(i)\|_2^2}. \quad (10)$$

Based on the recursion in (9), the convergence characteristics of the algorithm can be described. For this purpose, it is necessary to define the autocorrelation matrices of $\tilde{\psi}_k(i)$ and $\tilde{\mathbf{w}}_k(i)$ as follows:

$$\tilde{\mathbf{P}}_k(i) \triangleq E\{\tilde{\psi}_k(i) \tilde{\psi}_k^T(i)\}, \quad (11)$$

$$\mathbf{P}_k(i) \triangleq E\{\tilde{\mathbf{w}}_k(i) \tilde{\mathbf{w}}_k^T(i)\}, \quad (12)$$

where $E[\cdot]$ represents the statistical expectation. Employing (9) in (11), it is established that

$$\begin{aligned} \tilde{\mathbf{P}}_k(i) &= E\{\mathbf{F}_k(i) \tilde{\mathbf{w}}_k(i-1) \tilde{\mathbf{w}}_k^T(i-1) \mathbf{F}_k^T(i)\} \\ &\quad - \eta_k E\left\{\mathbf{F}_k(i) \tilde{\mathbf{w}}_k(i-1) \frac{g(e_k(i)) v_k(i) \mathbf{u}_k(i)}{\|\mathbf{u}_k(i)\|_2^2}\right\} \\ &\quad - \eta_k E\left\{\frac{g(e_k(i)) v_k(i) \mathbf{u}_k(i)}{\|\mathbf{u}_k(i)\|_2^2} \tilde{\mathbf{w}}_k^T(i-1) \mathbf{F}_k^T(i)\right\} \\ &\quad + \eta_k^2 E\left\{\frac{g^2(e_k(i)) v_k^2(i) \mathbf{u}_k(i) \mathbf{u}_k^T(i)}{\|\mathbf{u}_k(i)\|_2^4}\right\}. \end{aligned} \quad (13)$$

To calculate the expectations in (13), the following assumptions are necessary:

Assumption 1: $\tilde{\mathbf{w}}_k(i)$ and $\mathbf{u}_k(i)$ are mutually independent, which is the commonly used independence assumption [20].

Assumption 2: equation (2) shows the value of $g(e_k(i))$ typically approaches 1 or 0 in absence or presence of impulsive noises, respectively. Hence, we can assume that $g(e_k(i))$ is weakly independent of the input regressor $\mathbf{u}_k(i)$ and the additive noise $v_k(i)$ [13].

Assumption 3: since the negative effect of impulsive noises is absorbed into $g(e_k(i))$, the additive noise $v_k(i)$ in (8) can be assumed to contain only the background noise $\theta_k(i)$, which is a zero-mean process with variance $\sigma_{\theta,k}^2$.

Assumption 4: When the adaptive filter length M is sufficiently large, the variation of the input signal energy between consecutive iterations becomes negligible [21], thus the approximation $E\{1/\|\mathbf{u}_k(i)\|_2^2\} \approx 1/E\{\|\mathbf{u}_k(i)\|_2^2\}$ is reasonable.

Based on Assumptions 1–4 and the lemma in [22], (13) can be simplified, which leads to the following MSD recursion:

$$\tilde{p}_k(i) = b_k(i) p_k(i-1) + t_k(i), \quad (14)$$

where

$$b_k(i) \triangleq 1 - \frac{2\eta_k E\{g(e_k(i))\} - \eta_k^2 E\{g^2(e_k(i))\}}{\delta M}. \quad (15)$$

$$t_k(i) \triangleq \frac{\eta_k^2 E\{g^2(e_k(i))\} \sigma_{\theta,k}^2}{E\{\|\mathbf{u}_k(i)\|_2^2\}}. \quad (16)$$

Note that $\tilde{p}_k(i) \triangleq \text{Tr}[\tilde{\mathbf{P}}_k(i)]$ denotes the intermediate MSD at node k , while $p_k(i) \triangleq \text{Tr}[\mathbf{P}_k(i)]$ represents the MSD at node k . Moreover, we take $\delta = 1$ for white inputs and $\delta > 1$ for correlated inputs [23].

Similarly, by subtracting $\mathbf{w}_o$ on both sides, the combination step (6b) results in

$$\tilde{\mathbf{w}}_k(i) = \sum_{l \in N_k} c_{l,k} \tilde{\mathbf{p}}_l(i). \quad (17)$$

By employing Jensen's inequality [24], after pre-multiplying both sides in (17) by their transposes and taking the expectation and the trace, we arrive at

$$p_k(i) \leq \sum_{l \in N_k} c_{l,k} \tilde{p}_l(i). \quad (18)$$

Remark 1: In (14), the term $b_k(i)$ governs the convergence behavior of the DNMCC algorithm. A larger step-size leads to faster convergence, while a smaller step-size results in slower convergence. On the other hand, the term $t_k(i)$ determines the steady-state performance of the DNMCC algorithm a larger step-size yields a lower/higher steady-state misadjustment.

To address the trade-off between convergence speed and steady-state misadjustment induced by the step-size, as discussed in Remark 1, the concept of a SSC is introduced, where the appropriate step-size that minimizes the predicted MSD trend of the DNMCC algorithm is adaptively selected at each iteration.

Although (14) provides the recursive expression for predicting the MSD trend, the presence of the background noise variance $\sigma_{\theta,k}^2$ and the expectation operators (i.e., $E\{g(e_k(i))\}$ and $E\{g^2(e_k(i))\}$) limits its practical applicability. To solve this problem, we propose two compensation measures:

- 1) In the steady-state, as $\tilde{\mathbf{w}}_k(i) \rightarrow 0$, the nodes' error becomes $e_k(i) \approx v_k(i)$. Therefore, the energy of $e_k^2(i)$ excluding impulsive noises can be used to approximate the background noise variance $\sigma_{\theta,k}^2$ by employing a smoothing recursion given by

$$h_{k,1}(i) = \lambda h_{k,1}(i-1) + (1-\lambda)g^2(e_k(i))e_k^2(i), \quad (19a)$$

where $\lambda \in (0.9, 1)$ is a forgetting factor.

- 2) Since the direct computation of $E\{g(e_k(i))\}$ and $E\{g^2(e_k(i))\}$ is not feasible, we also employ a smoothing technique to recursively estimate the statistical quantities required for the MSD prediction:

$$h_{k,2}(i) = \lambda h_{k,2}(i-1) + (1-\lambda)g(e_k(i)), \quad (19b)$$

$$h_{k,3}(i) = \lambda h_{k,3}(i-1) + (1-\lambda)g^2(e_k(i)). \quad (19c)$$

Moreover, according to Assumption 4, the expectation operation over $\mathbf{u}_k(i)$ in (14) is approximated by its instantaneous realization. As a result of the above measures, (14) can be rewritten as

$$\begin{aligned} \tilde{p}_k(i) = & \left(1 - \frac{2\eta_{k,s}h_{k,2}(i) - \eta_{k,s}^2h_{k,3}(i)}{\delta M}\right) p_k(i-1) \\ & + \frac{\eta_{k,s}^2h_{k,1}(i)}{\|\mathbf{u}_k(i)\|_2^2}. \end{aligned} \quad (20)$$

To enable the adaptive step-size selection, a descending set of candidate step-sizes for node k is introduced:

$$\boldsymbol{\eta}_k = \{\eta_{k,1}, \eta_{k,2}, \dots, \eta_{k,S}\}, \quad (21)$$

where S denotes the total number of the step-size values and $s \in \{1, 2, \dots, S\}$ is the index of the candidate step-size. For each candidate $\eta_{k,s}$, an individual MSD trend can be predicted according to

$$\begin{aligned} \tilde{p}_{k,s}(i) = & \left(1 - \frac{2\eta_{k,s}h_{k,2}(i) - \eta_{k,s}^2h_{k,3}(i)}{\delta M}\right) p_k(i-1) \\ & + \frac{\eta_{k,s}^2h_{k,1}(i)}{\|\mathbf{u}_k(i)\|_2^2}. \end{aligned} \quad (22)$$

At each iteration, node k selects the appropriate step-size that minimizes its MSD trend, that is,

$$\eta_{k,s_o}(i) = \arg \min_{\eta_{k,s}} \{\tilde{p}_{k,s}(i)\}_{s=1}^S. \quad (23)$$

The predicted MSD corresponding to the selected step-size $\eta_{k,s_o}(i)$ is represented as $\tilde{p}_{k,s_o}(i)$.

Following the combination rule in (6b), each node fuses the local minimum $\tilde{p}_{k,s_o}(i)$ values from its neighbors as

$$p_k(i) = \sum_{l \in N_k} c_{l,k} \tilde{p}_{l,s_o}(i). \quad (24)$$

Since the SSC strategy only compares the evolution trends of the MSD associated with different step-sizes from the same starting point, rather than pursuing an accurate MSD value, the specific choice of the initial value $p_k(0)$ is not critical. Hence, it can be simply set to a constant larger than one.

IV. SIMULATION RESULTS

To evaluate the performance of the proposed SSC-DNMCC algorithm, we consider a distributed network with 20 nodes, as shown in Fig. 1. The unknown vector $\mathbf{w}_o$ is randomly generated from a zero-mean uniform distribution, has length $M = 128$, and is normalized to satisfy $\|\mathbf{w}_o\|_2 = 1$. At node k , the input vector is $\mathbf{u}_k(i) = [u_k(i), u_k(i-1), \dots, u_k(i-M+1)]^T$, where $u_k(i)$ denotes the i -th sample of a zero-mean white Gaussian process with variance $\sigma_{\epsilon,k}^2$, unless stated otherwise. The variances $\sigma_{\epsilon,k}^2$ and $\sigma_{\theta,k}^2$ for all nodes are provided in Fig. 2. For all diffusion-type algorithms, the combination coefficients $\{c_{l,k}\}$ follow the Metropolis rule [25]. Algorithm performance is assessed through the network mean-square deviation (MSD), defined as

$$\text{MSD}(i) = \frac{1}{N} \sum_{k=1}^N \|\mathbf{w}_k(i) - \mathbf{w}_o\|_2^2. \quad (25)$$

All simulation curves are averaged over 100 independent trials.

At node k , in addition to the background noise $\theta_k(i)$, the additive noise $v_k(i)$ includes impulsive noise $\xi_k(i)$ modeled by a Bernoulli-Gaussian (BG) process [21], which is defined as $\xi_k(i) = \tau_k(i)f_k(i)$. Here, $\tau_k(i)$ is a Bernoulli random variable with $\Pr[\tau_k(i) = 1] = q_{r,k}$ and $\Pr[\tau_k(i) = 0] = 1 - q_{r,k}$, while $f_k(i)$ is a zero-mean white Gaussian process with variance $\sigma_{f,k}^2 = \hbar\sigma_{\theta,k}^2$, where $\hbar \gg 1$. The parameter $q_{r,k}$ denotes the occurrence probability of impulsive noises, and we set $q_{r,k} = 0.01$ and $\hbar = 10000$ in our simulations.

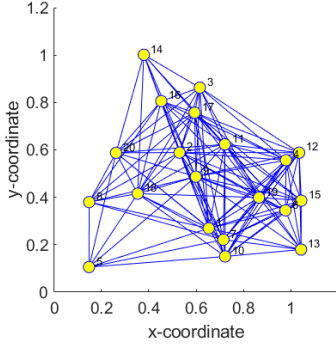


Fig. 1: Network topology with $N = 20$ nodes.

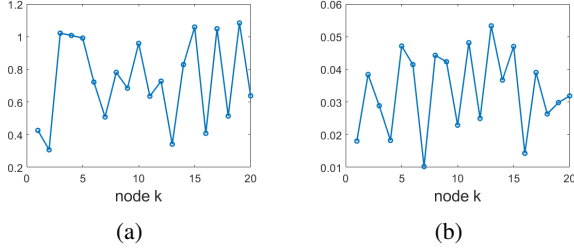


Fig. 2: Values of (a) $\sigma_{\epsilon,k}^2$ and (b) $\sigma_{\theta,k}^2$ at different nodes

Fig. 3 illustrates the performance comparison of the DNMCC and SSC-DNMCC algorithms. As observed, larger step-sizes lead to faster convergence and higher steady-state misadjustment, while smaller step-sizes yield slower convergence and lower steady-state misadjustment. In contrast, due to the use of the proposed SSC mechanism, the SSC-DNMCC algorithm adaptively tunes the step-size during different learning phases, enabling it to achieve both rapid convergence and low steady-state misadjustment.

Fig. 4 compares the performance of the DNLMS, VSS-

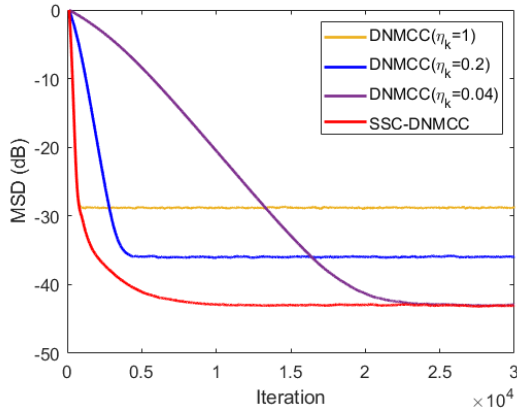


Fig. 3: Network MSD curves of the DNMCC and SSC-DNMCC algorithms. Parameter setting of the SSC-DNMCC algorithm is: (a) $p_k(0) = 10$, $S = 3$, $\eta_k = [1, 0.2, 0.04]$, $\delta = 1$, $\lambda = 0.99$.

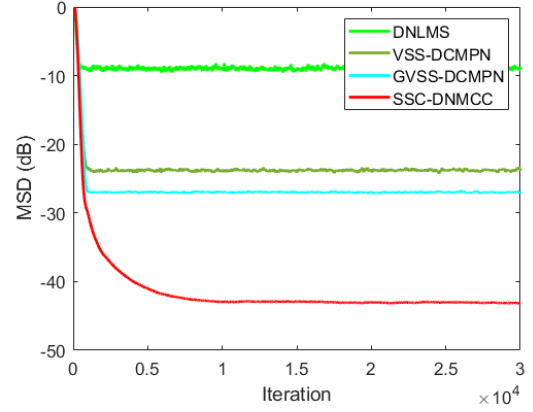


Fig. 4: Network MSD curves of various diffusion algorithms. Parameter setting of algorithms is as follows: $\mu = 0.9$ (DNLMS); $\mu = 0.003$, $\kappa = 0.01$ (VSS-DCMPN); $\mu = 0.003$, $\kappa = 0.01$, $\theta = -2$ (GVSS-DCMPN). Parameter setting of SSC-DNMCC is the same as in Fig. 3.

DCMPN, GVSS-DCMPN, and proposed SSC-DNMCC algorithms. Among them, the DNLMS algorithm exhibits the poorest performance due to its inability to suppress impulsive noise. In contrast, the other algorithms demonstrate strong robustness against impulsive noises. Importantly, the proposed SSC-DNMCC algorithm achieves a lower steady-state misadjustment while maintaining a convergence speed comparable with that of VSS-DCMPN and GVSS-DCMPN algorithms.

V. CONCLUSIONS

In this study, a normalized structure was introduced to mitigate the effect of highly non-uniform input signal powers on the DMCC algorithm. To resolve the trade-off between convergence speed and steady-state performance, an SSC mechanism was incorporated, leading to the SSC-DNMCC algorithm. The proposed SSC approach established a practical expression of the MSD trend and selected the optimal step size from a predefined set to minimize this trend. Simulation results verified that the proposed algorithms achieved significant performance gains over other robust diffusion filtering methods.

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